SHOPSPHERE ENGINEERING
INCIDENT REPORT

Initial Incident Report: INC-2026-1042

Elevated payment failures after 10:05 UTC on 2026-10-23. First responder report.

Incident ID	INC-2026-1042
Report written	2026-10-23 10:18 UTC
Author	Priya Nair, platform engineer (first responder)
Status	Investigating
Severity	SEV-2 (customer payments failing)
Services	payment-service, order-service

Fictional document. ShopSphere, its services, people and data are synthetic. This document is part of the TraceIQ training dataset and is not documentation for any real system.

Contents

Incident Details . . . 3
1. Summary . . . 3
2. Impact So Far . . . 3
3. Suspected Cause . . . 4
4. Actions Taken . . . 4
5. Next Steps . . . 4
6. Open Questions . . . 5
7. Recent Change: payment-service 2.4.0 Deploy . . . 5
8. Timeline So Far . . . 5
9. Evidence Collected . . . 6
9.1 Alert . . . 6
9.2 Dashboard Snapshot . . . 6
9.3 Error Screenshot . . . 7
10. Services Involved . . . 7
11. Communication Log . . . 8
12. Not Yet Checked . . . 8
13. Glossary . . . 8
Appendix A. Metric Rows Used . . . 9
Appendix B. Error Screenshot Text . . . 10
Appendix C. Deployment Events Used . . . 10

Incident Details

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Incident metadata.

> **Early report.** This report was written at 10:18 UTC, early in the incident. Statements in Section 3 are a first guess and are marked as not verified.

1. Summary

Since about 10:05 UTC, most checkout payments are failing. payment-service returns error code PAY-401. The PaymentFailureRateHigh alert fired at 10:09 UTC. payment-service version 2.4.0 was deployed shortly before (around 10:02 UTC).

Question	Answer at 10:18 UTC
What is failing?	Most checkout payments.
Since when?	About 10:05 UTC.
Which error?	payment-service returns error code PAY-401.
Alert	PaymentFailureRateHigh fired at 10:09 UTC.
Recent change	payment-service 2.4.0 deployed shortly before (around 10:02 UTC).

2. Impact So Far

- Roughly 3 out of 4 orders end in PAYMENT_FAILED.
- Some payments still succeed. Not yet known which ones.
- Customers see a generic "payment could not be completed" message.

The order-level numbers from the last complete 5-minute buckets support the first point:

Bucket start (UTC)	Orders	Orders with failed payment	Failed (%)
09:55	571	0	0.0
10:00	625	0	0.0
10:05	614	485	79.0
10:10	571	445	77.9

Table 2-1. order-service orders and failed payments (metrics rows M-012, M-014, M-016, M-018).

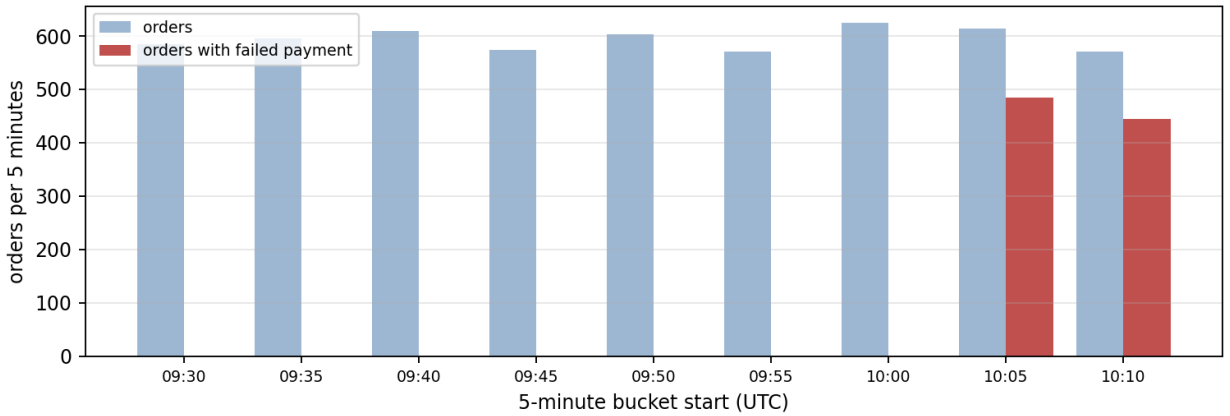


Figure 2-1. Orders and orders with failed payment per 5-minute bucket, 09:30 to the 10:10 bucket (order-service metrics).

3. Suspected Cause

PAY-401 looks like an unauthorized error. My first guess is an authentication or credential problem between order-service and payment-service, for example an expired or rotated API key that 2.4.0 now checks more strictly. This is not verified yet.

Hypothesis	Status at 10:18 UTC	How to check
Authentication or credential problem between order-service and payment-service (for example an expired or rotated API key that 2.4.0 now checks more strictly).	Not verified	Check when the order-service to payment-service API key was last rotated.
Failures are linked to the 2.4.0 deploy.	Not verified	Check whether failures are linked to the 2.4.0 deploy.

Table 3-1. Open hypotheses at the time of writing.

4. Actions Taken

- Opened the incident at 10:15 UTC and paged the payments and order-service owners.
- Started an incident call for 10:20 UTC.

5. Next Steps

1. Check when the order-service to payment-service API key was last rotated.
2. Check whether failures are linked to the 2.4.0 deploy.
3. Decide whether to roll back.

6. Open Questions

- Why do some payments still succeed?
- Is the latency increase on the order-service dashboard related?

7. Recent Change: payment-service 2.4.0 Deploy

The only production change found so far near the start of the failures is the payment-service deploy below. Whether it is linked to the failures is not verified yet (Next Steps, item 2).

Item	Value
Service	payment-service
Environment	production
From version	2.3.1
To version	2.4.0
Deploy pipeline run	PIPE-7781
Deploy started	2026-10-23T10:02:00Z
Deploy completed	2026-10-23T10:04:30Z (All 4 instances running 2.4.0.)
Health check	2026-10-23T10:05:00Z: GET /healthz returned 200 on 4/4 instances. Health check does not send payment requests.

Table 7-1. Deploy details from the deployment events (EVT-001, EVT-002, EVT-003).

Important. The health check passed at 10:05:00, but it does not send payment requests. A passing health check does not show that payments work.

8. Timeline So Far

All times UTC on 2026-10-23. Entries come from deployment events, the alert, and the incident record.

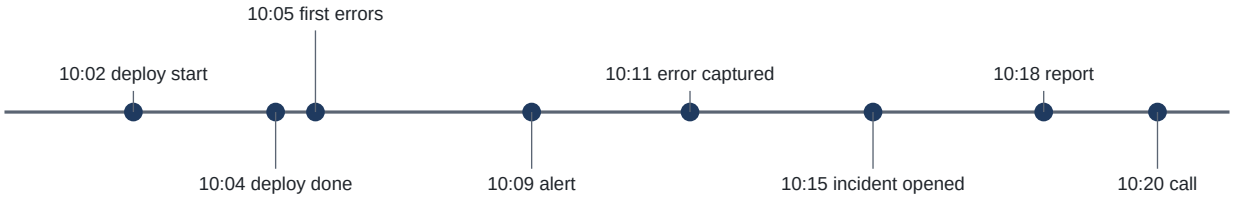


Figure 8-1. Events from 10:00 to 10:20 UTC (not all at exact scale for labels).

Time	Event	Details
10:02:00	deploy started	Rolling deploy of 4 instances started by deploy pipeline run PIPE-7781.
10:04:30	deploy completed	All 4 instances running 2.4.0.
10:05:00	health check	GET /healthz returned 200 on 4/4 instances. Health check does not send payment requests.
~10:05	first errors	Payments start failing with error code PAY-401.
10:09:00	alert triggered	PaymentFailureRateHigh: failure rate above 5% for 5 minutes.
10:11:47	error captured	order-service debug console captures a 422 response with PAY-401 for request req-1107 (Figure 9-2).
10:15:00	incident opened	Incident opened for elevated payment failures.
10:18	report written	This initial report.
10:20	incident call	Call scheduled.

Table 8-1. Timeline up to the time of writing.

9. Evidence Collected

9.1 Alert

Item	Value
Alert	PaymentFailureRateHigh
Fired at	10:09 UTC
Condition	Payment failure rate above 5% for 5 minutes

9.2 Dashboard Snapshot

Failure rate and p95 latency per 5-minute bucket, from 09:30 to the 10:10 bucket (the last complete bucket at the time of writing).

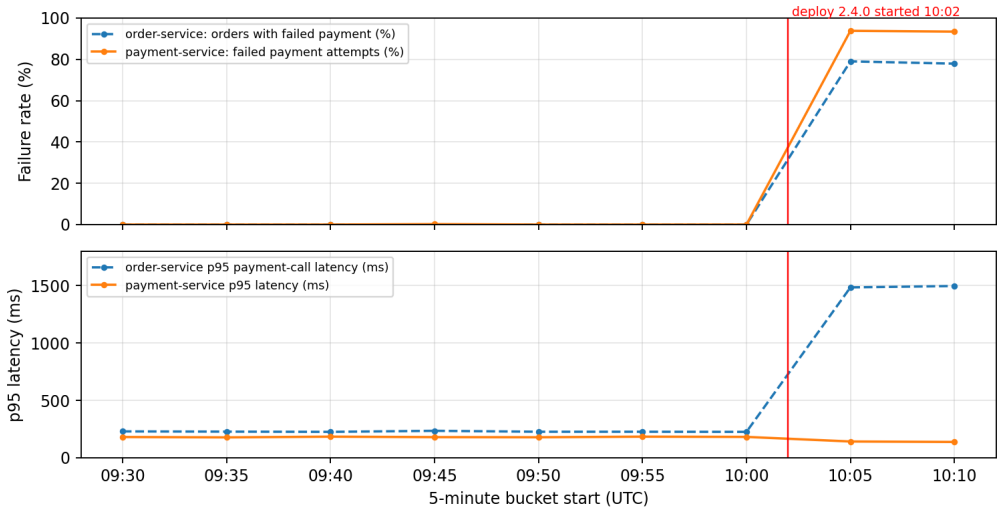


Figure 9-1. Payment failure rate and p95 latency up to the 10:10 bucket.

Bucket (UTC)	Service	Requests	Errors	Error %	p95 ms
09:55	order-service	571	0	0.0	227
09:55	payment-service	571	0	0.0	184
10:00	order-service	625	0	0.0	226
10:00	payment-service	625	0	0.0	182
10:05	order-service	614	485	79.0	1485
10:05	payment-service	2069	1940	93.8	142
10:10	order-service	571	445	77.9	1497
10:10	payment-service	1906	1780	93.4	138

Table 9-1. Metric rows behind Figure 9-1 (payment-service counts every attempt; order-service counts orders).

9.3 Error Screenshot



Figure 9-2. order-service debug console, 10:11:47 UTC, request req-1107, trace trc-5a10c4.

10. Services Involved

The failing call is the payment call from order-service to payment-service.

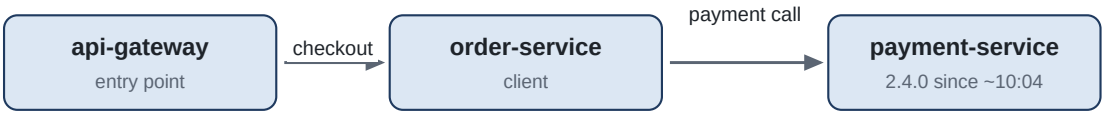


Figure 10-1. Request path of a checkout payment.

11. Communication Log

Time (UTC)	Action
10:15	Incident INC-2026-1042 opened. Payments and order-service owners paged.
10:18	Initial report written (this document).
10:20	Incident call starts.

12. Not Yet Checked

At 10:18 UTC the following had not been looked at. They are listed so the incident call can assign them.

Item	Why it may matter
When the order-service to payment-service API key was last rotated	Tests the suspected cause in Section 3.
Whether failures are linked to the 2.4.0 deploy	Failures began shortly after the deploy.
Which payments still succeed	Open question 1 in Section 6.
Why order-service latency rose	Open question 2 in Section 6.
Data stores (redis, postgres)	Not checked yet.
Whether to roll back	Next Steps, item 3.

Table 12-1. Open items.

Failed orders in the two complete buckets after the first errors: 930 orders (10:05 and 10:10 buckets, order-service metrics). The number of affected customers has not been calculated.

13. Glossary

Term	Meaning
PAY-401	Error code returned by payment-service in this incident.
PAYMENT_FAILED	Order outcome when payment does not succeed.
PaymentFailureRateHigh	Alert for payment failure rate above 5% for 5 minutes.
SEV-2	Severity level used for this incident: customer payments failing.
Trace ID	ID that links a response to log records (for example trc-5a10c4).

Appendix A. Metric Rows Used

All metric rows from 09:30 up to the 10:10 bucket, as exported every 5 minutes. payment-service rows count every payment attempt; order-service rows count orders.

Row	Bucket	Service	Version	Req.	Errors	Err. %	p50	p95	p99
M-001	09:30	payment-service	2.3.1	585	0	0.0	85	181	255
M-002	09:30	order-service	5.7.0	585	0	0.0	140	230	325
M-003	09:35	payment-service	2.3.1	595	0	0.0	86	178	262
M-004	09:35	order-service	5.7.0	595	0	0.0	141	228	323
M-005	09:40	payment-service	2.3.1	610	0	0.0	85	184	262
M-006	09:40	order-service	5.7.0	610	0	0.0	140	226	331
M-007	09:45	payment-service	2.3.1	575	1	0.2	86	180	3010
M-008	09:45	order-service	5.7.0	574	0	0.0	141	235	3290
M-009	09:50	payment-service	2.3.1	603	0	0.0	85	179	258
M-010	09:50	order-service	5.7.0	603	0	0.0	141	227	320
M-011	09:55	payment-service	2.3.1	571	0	0.0	86	184	261
M-012	09:55	order-service	5.7.0	571	0	0.0	142	227	320
M-013	10:00	payment-service	mixed	625	0	0.0	86	182	267
M-014	10:00	order-service	5.7.0	625	0	0.0	138	226	326
M-015	10:05	payment-service	2.4.0	2069	1940	93.8	14	142	235
M-016	10:05	order-service	5.7.0	614	485	79.0	1441	1485	1558
M-017	10:10	payment-service	2.4.0	1906	1780	93.4	13	138	234
M-018	10:10	order-service	5.7.0	571	445	77.9	1450	1497	1558

Table A-1. metrics.csv rows M-001 to M-018. Latency in ms.

Appendix B. Error Screenshot Text

Text visible in Figure 9-2, copied for searching. The request body was hidden by the console privacy setting.

Figure 9-2 as text

```
order-service debug console  |  production  |  SYNTHETIC DATA
2026-10-23T10:11:47.203Z  request req-1107  (order ORD-88251)
> POST http://payment-service/v2/payments
> Content-Type: application/json
> Idempotency-Key: idem-ORD-88251

< HTTP/1.1 422 Unprocessable Entity
< Content-Type: application/json
<
{
  "error_code": "PAY-401",
  "message": "Request validation failed",
  "trace_id": "trc-5a10c4"
}

retry 1 of 3 scheduled in 200 ms ...  (request body hidden by console privacy setting)
```

Appendix C. Deployment Events Used

Deployment and incident events recorded up to the time of writing (deployment_events.json).

event_id timestamp event_type old -> new status

```
EVT-001  2026-10-23T10:02:00Z  deploy_started   2.3.1 -> 2.4.0  in_progress
EVT-002  2026-10-23T10:04:30Z  deploy_completed 2.3.1 -> 2.4.0  succeeded
EVT-003  2026-10-23T10:05:00Z  health_check     2.4.0 -> 2.4.0  passed
EVT-004  2026-10-23T10:09:00Z  alert_triggered  2.4.0 -> 2.4.0  firing
EVT-005  2026-10-23T10:15:00Z  incident_opened  2.4.0 -> 2.4.0  open
```

Event	Note
EVT-001	Rolling deploy of 4 instances started by deploy pipeline run PIPE-7781.
EVT-002	All 4 instances running 2.4.0.
EVT-003	GET /healthz returned 200 on 4/4 instances. Health check does not send payment requests.
EVT-004	PaymentFailureRateHigh: failure rate above 5% for 5 minutes.
EVT-005	Incident opened for elevated payment failures.

Table C-1. Event notes.